PRANJAL KARAN

Computer Science Engineering Student

+91-9162084158 — pranjalkaran2004@gmail.com — pranjalk.vercel.app — linkedin.com/in/pranjal-karan — github.com/Pranjal-k22

Education

Vellore Institute of Technology – Bachelor of Technology, Computer Science & Engineering **Sept 2023 – May 2027**

CGPA: 8.70/10

Technical Skills

Languages: JavaScript (ES6+), C++, SQL

Backend & APIs: Node.js, Express.js, RESTful API Design, JWT Authentication, Clerk Auth

Frontend: React, Tailwind CSS v4

Databases: MongoDB, MySQL, PostgreSQL

Web & Networking: HTTP/HTTPS, DNS, TLS, CORS, Network Debugging

Engineering & DevOps: Git, GitHub, VS Code

Integrations & AI: API Integration

Projects

ReconAI – Verification-First AI Finance Controller Platform — GitHub **2026** React, Vite, Tailwind CSS, Node.js, Express.js, MongoDB

- **Built** a full-stack financial operations platform that reconciles merchant orders, gateway payments, and settlement records using an **integer-paise deterministic engine**, processing a **120-scenario benchmark** with **100% classification accuracy**.
- **Engineered** an **autonomous 9-state AI Finance Controller Agent** that orchestrates multi-pass candidate matching, confidence safety gates, exception creation, and system-triggered AI investigation dispatch.
- **Integrated Razorpay Test Mode** as a read-only API source with PII stripping, idempotent bulk upserts, and live-key guards; added **Google Gemini** for advisory root-cause analysis with a deterministic fallback.
- **Ensured** system reliability with **181 automated backend tests across 42 files**, enforcing append-only audit trails, credential redaction, and strict financial safety boundaries.

HealthPulse – Full-Stack Healthcare Appointment & Clinical Platform — Live — GitHub **2026** React 18, TypeScript, Node.js, Express.js, MongoDB

- **Architected** a highly scalable platform featuring **61 REST API endpoints** across 9 modules and **13 MongoDB schemas**, implementing a strict layered **controller-service-model architecture**.
- **Eliminated** double-booking race conditions under concurrent load by combining a **database-level compound partial unique index** with **TTL-based 5-minute slot holds**, guaranteeing atomic data integrity.
- **Engineered** stateless JWT authentication with a `tokenVersion` mechanism enabling instant session revocation, role-based access control across 3 personas, and rate-limited endpoints to prevent brute-force attacks.
- **Orchestrated** a dual-engine AI clinical-summary pipeline (Google Gemini + Ollama via `Promise.allSettled`) with a zero-hallucination medication guard, backed by **154 regression tests** and a **98/100** Lighthouse score.

Certifications

- **Microsoft Certified: Azure AI Cloud Developer Associate** (Score: 922/1000 — ID: DDFC0151CEFD415C)
- **Microsoft Certified: Azure Data Fundamentals** (Score: 955/1000)
- **MERN Full Stack Certification** (Ethnus & Codemithra)

Achievements & Leadership

- **National Qualifier** – Cleared Round 1 and advanced in **TCS CodeVita Season 13**, a highly competitive national-level programming challenge.
- **Social Media Co-Lead, Linpack Club** (VIT Bhopal) – Directed a core team to design and execute digital promotional campaigns for university-wide technical events from Sept 2024 to Feb 2026. Managed content strategy and outreach, successfully increasing overall student engagement and event turnout by **30%**.